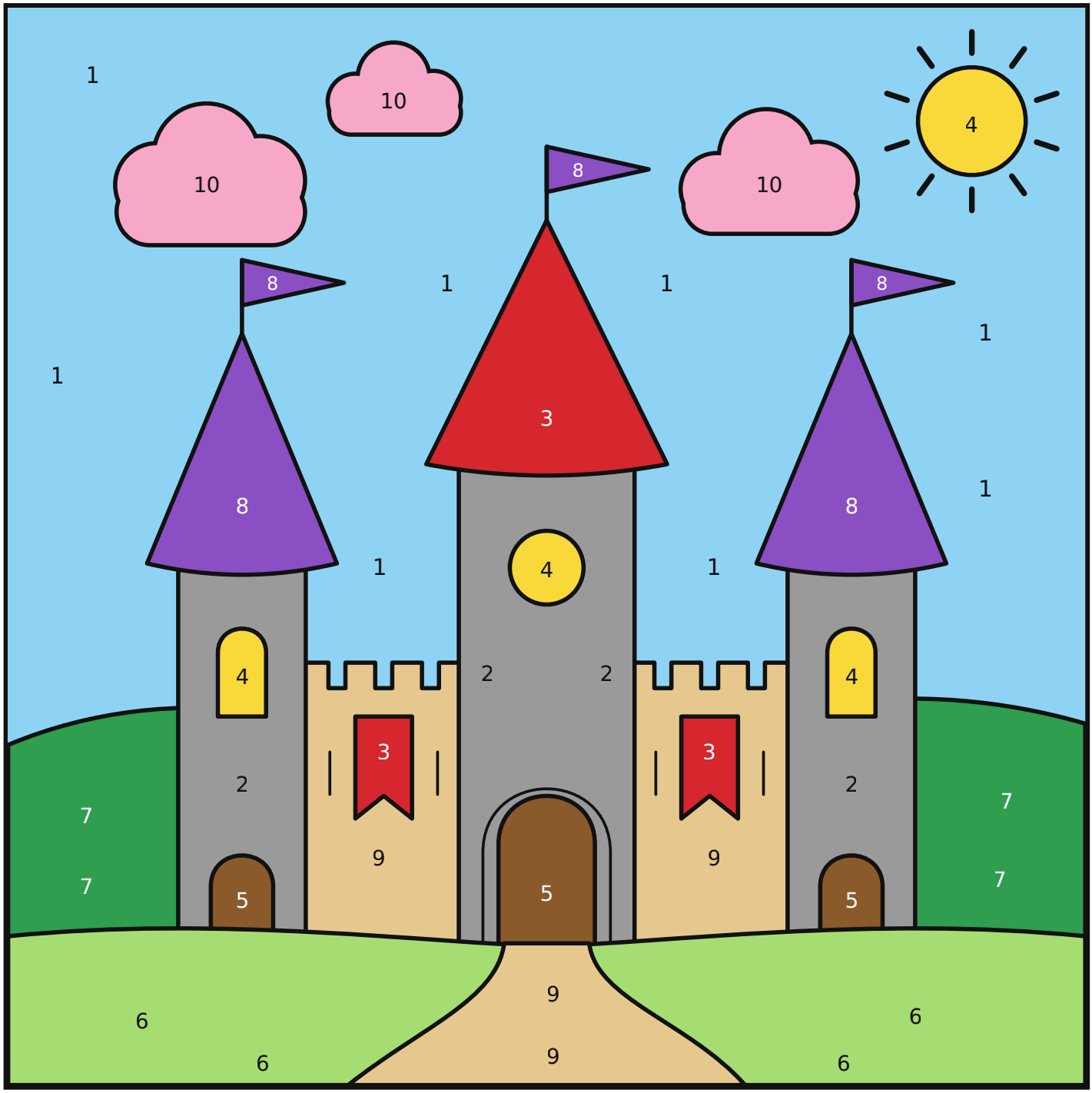
Name: **ANSWER KEY** _____ Date: _____

Polynomials & Factoring: Color By Number



Color Code:

5 Brown $(x - 3)(x + 3)$	**7** Green $(2x + 1)(x + 3)$	**6** Light Green $2x(x + 4)$	**2** Gray $(x - 2)(x - 3)$	**3** Red $(x - 2)(x + 3)$
10 Pink $3(x - 2)(x + 2)$	**9** Tan $x^2 - 3x - 10$	**8** Purple $(x - 3)^2$	**4** Yellow $(x + 2)(x - 3)$	**1** Light Blue $(x + 2)(x + 3)$

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Polynomials & Factoring: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Factor: $x^2 + 5x + 6$

find two numbers: product 6, sum 5 → 2 and 3
$(x + 2)(x + 3)$

1. Color: **Light Blue**

2.
Factor: $x^2 - 5x + 6$

product 6, sum −5 → −2 and −3
$(x - 2)(x - 3)$

2. Color: **Gray**

3.
Factor: $x^2 + x - 6$

product −6, sum 1 → 3 and −2
$(x - 2)(x + 3)$

3. Color: **Red**

4.
Factor: $x^2 - x - 6$

product −6, sum −1 → −3 and 2
$(x + 2)(x - 3)$

4. Color: **Yellow**

5.
Factor: $x^2 - 9$

difference of squares: $a^2 - b^2 = (a - b)(a + b)$
$(x - 3)(x + 3)$

5. Color: **Brown**

6.
Factor completely: $2x^2 + 8x$

GCF = 2x
$2x(x + 4)$

6. Color: **Light Green**

7.
Factor: $2x^2 + 7x + 3$

multiply $2 \cdot 3 = 6$; find two numbers with sum 7: 1 and 6
split the middle: $2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1)$
$(2x + 1)(x + 3)$

7. Color: **Green**

8.
Factor: $x^2 - 6x + 9$

perfect square trinomial: $(x - 3)(x - 3)$
$(x - 3)^2$

8. Color: **Purple**

9.
Multiply: $(x + 2)(x - 5)$

$x^2 - 5x + 2x - 10$
$x^2 - 3x - 10$

9. Color: **Tan**

10.
Factor completely: $3x^2 - 12$

GCF = 3: $3(x^2 - 4)$
$3(x - 2)(x + 2)$

10. Color: **Pink**